

In 47–50 (a) use the logical equivalences $p \rightarrow q \equiv \sim p \vee q$ and $p \leftrightarrow q \equiv (\sim p \vee q) \wedge (\sim q \vee p)$ to rewrite the given statement forms without using the symbol $\rightarrow$ or $\leftrightarrow$, and (b) use the logical equivalence $p \vee q \equiv \sim(\sim p \wedge \sim q)$ to rewrite each statement form using only $\wedge$ and $\sim$.

48. $p \vee \sim q \rightarrow r \vee q$

(a) $p \vee \sim q \rightarrow r \vee q$

$$(p \vee \sim q) \rightarrow (r \vee q) \equiv \sim(p \vee \sim q) \vee (r \vee q)$$

$$\sim(p \vee \sim q) \vee (r \vee q) \equiv \sim p \wedge q \vee r \vee q$$

(b) $\sim p \wedge q \vee r \vee q$

$$(\sim p \wedge q) \vee (r \vee q) \equiv \sim(\sim(\sim p \wedge q) \wedge \sim(r \vee q))$$

$$\sim(\sim(\sim p \wedge q) \wedge \sim(r \vee q)) \equiv \sim((p \vee \sim q) \wedge (\sim r \wedge \sim q))$$

$$\sim((p \vee \sim q) \wedge (\sim r \wedge \sim q))$$

The assignment was to rewrite only using $\wedge$ and $\sim$. So let's now focus on the only part of the statement form which still contains $\vee$:

$$(p \vee \sim q) \equiv \sim(\sim p \wedge \sim(\sim q))$$

$$\sim(\sim p \wedge \sim(\sim q)) \equiv \sim(\sim p \wedge q)$$

We can now insert the statement $\sim(\sim p \wedge q)$ back into the statement form:

$$\sim(\sim(\sim p \wedge q) \wedge (\sim r \wedge \sim q))$$

Solution: (a) $p \vee \sim q \rightarrow r \vee q \equiv \sim p \wedge q \vee r \vee q$
 (b) $\sim(\sim(\sim p \wedge q) \wedge (\sim r \wedge \sim q))$